

# Mobile Technology (How to use Medical Software Programs)

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# Outline

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Learning Outcomes

Introduction

Need for Mobile Devices at the Point of Care

Basic Organizer Functions

Medical Software Programs

Limitations of Mobile Technology

# Learning Outcomes

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List features of medical mobile technology

Describe medical mobile technology and its limitations

# Introduction

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Mobile technology (examples: personal digital assistant, smartphones) is a very logical transitional step from the personal computer. With improving speed, memory, wireless connectivity and shrinking size, consumers desire a more portable platform for their information and applications.

# Need for Mobile Devices at the Point of Care

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One major **motivation** driving the widespread adoption of mobile devices by HCPs (Health care professionals) has been the need for **better communication** and **information resources** at the point of care.

HCPs require access to many **types of resources** in a clinical setting, including:

- Communication capabilities—voice calling, video conferencing, text, and e-mail.
- Hospital information systems (HISs)—electronic health records (EHRs), electronic medical records (EMRs), clinical decision support systems (CDSSs), picture archiving and communication systems (PACSs), and laboratory information systems (LISs).
- Informational resources—textbooks, guidelines, medical literature, drug references.
- Clinical software applications—disease diagnosis aids, medical calculators.

# Mobile Technology Popularity

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Many medical schools fully support **handheld technology** as part of their educational programs.

Multiple studies have shown, a **drug reference** is listed as the most frequently used program.

Personal scheduling is listed second because most people need to keep track of their schedules and most programs **synchronize** with Microsoft Outlook, Google OneDrive.



# Basic Organizer Functions



All smartphones have basic organizer functions such as contacts, calendars and the ability to record memos, tasks and store passwords. This allows medical workers to synchronize their work calendar to their smartphones. Google mobile has the feature of synching its calendar feature to multiple smartphones running the Android, Mac OS X, Windows Mobile, Symbian and BlackBerry operating systems.

With the evolution to cloud computing, more and more medical programs will be hosted on the **cloud** and not on the device.



# How HCPs Use Mobile Devices and Apps

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Health care professionals (HCPs) use **medical devices and apps** for **many purposes**: administration (information management and time management), health record maintenance and access, communications and consulting, reference and information gathering, clinical decision making, patient monitoring and medical education and training.



# MOBILE APPS FOR HEALTH CARE PROFESSIONALS

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## What Are “Apps”?

Apps are software programs that have been developed to run on a computer or mobile device to accomplish a specific purpose.

Faster processors, improved memory, smaller batteries, and highly efficient open-source operating systems that perform complex functions have paved the way for the development of a flood of medical mobile device apps for both professional and personal use.

The ability to download medical apps on mobile devices has made a wealth of mobile clinical resources available to HCPs.

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There are even mobile apps that simulate surgical procedures or that can conduct simple medical exams, such as hearing or vision tests.

**Apple created the “Apps for Healthcare Professionals”** section within the medical category of the iTunes Appstore, a unique feature among mobile app marketplaces.

“Google Play” shop that provides a wide variety of apps, including some for HCPs, for mobile devices that use the Android operating system.



# Textbooks for Mobile Technology

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Trying to read a textbook on a small screen will not appeal to everyone. Nevertheless, there are many textbooks available for use on handheld technology. As a result of secure digital cards many textbooks can be stored on a memory card, thus saving space on the internal handheld memory.

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Many well-known medical journals and medical textbooks can be purchased as mobile apps after payment of a subscription fee. Medical textbook apps are often updated annually, eliminating the need to buy newer editions.

# Medical apps for health care professionals

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**Information Management:** Evernote, Notability

**Communication and Consulting:** Doximity

**Reference and Information Gathering:** Epocrates, Skyscape, Medscape

**Patient Management and Monitoring:** Archimedes

**Medical Education and Training:** QuantiaMD

# Medical apps for health care professionals conti...(General medical software)

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Lexi-Comp: offers a variety of drug information references and decision support tools to meet the unique needs of pharmacists.

MobiPocket: Allows the reader to read e-books. Another company that sells medical e-books that require a special reader downloaded to the smartphone.

Skyscape: One of the most popular sites for e-textbooks for mobile technology is Skyscape. They offer over 300 titles covering primary care and multiple specialties.

# Medical Software Programs: Calculators

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Cardiac Clearance. A free program, for cardiac clearance before surgery; Simply tap on a few screens to answer standard questions and it tells you if the patient needs further testing prior to going to the operating room.

Cholesterol: To calculate actual risk for coronary heart disease.

Depression: A free program that uses a standard depression questionnaire and calculates the degree of depression.

Growth Charts. A free program that inputs the age, gender, height and weight and plots how the child compares to normative data

# Operating Systems for Smartphones

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- BlackBerry
- Windows
- IOS
- Android





# Few Major Technologies that are making mobile services a reality

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Global System for Mobile Communication (GSM)

Short Message Service (SMS)

Wireless Application Protocol (WAP)

Bluetooth

Wireless Markup Language (WML)

Wi-Fi (stands for wireless fidelity)

# Examples of Saudi medical Apps

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The "Tabaud" application

“Sehaty”

Tawakkalna



# Examples of Saudi medical Apps

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**The "Tabaud" application** is a way to notify those who are in contact with people infected with coronavirus.

**"Sehaty"** : It is the unified platform of Ministry of Health, which allows users to access health information and obtain several health services provided by various entities in the health sector in Saudi Arabia.

**Tawakkalna:** It is the **super app** that unifies government services in one platform, organizing services, information, documents, and posts to make them easier to use and improve the quality of life in Saudi Arabia in line with the goals of Vision 2030.

# Limitations of Mobile Technology

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1. Slow inputting: typical keyboards are too small and inputting with styluses is too slow.
2. Small screen: traditionally, handheld computers had tiny screens. Several phones like the iPhone have larger screens to make viewing of web pages and documents more reasonable.

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3. Security: as smartphones are used to access enterprise or patient level data, better security will need to be standard. Encryption and anti-viral programs for the mobile platform exist.
  4. Expense: not everyone is willing to pay for a data plan and other bells and whistles associated with smartphones.
  5. Adoption: It seems likely that most professionals will be using smartphones for professional and personal reasons in the near future.

# References

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Ventola, C.L., 2014. Mobile devices and apps for health care professionals: uses and benefits. *Pharmacy and Therapeutics*, 39(5), p.356.

